Darren Lee

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Education

University of California, Berkeley Expected Graduation: May 2029

B.S. Electrical Engineering and Computer Science, Minor in Materials Science and Engineering *Berkeley, CA*

- **Relevant Coursework:** Computer Architecture, Circuits and Devices, Properties of Materials, Signals and Systems, Data Structures, Linear Algebra + Diff Eq, Physics for Scientists and Engineers, SICP

Ruben S. Ayala High School *Chino Hills, CA* May 2025

- GPA: 4.61, ACT: 35, 16 AP Classes (All 5's and 4's), Class Rank: 7/672
- **Relevant Coursework:** AP Calc AB/BC, Stats, Physics 1 and C Mech/E&M, CSA, CSP

Skills

Hardware: Circuit design, Circuit simulation (LTspice), PCB design (KiCad), Embedded systems design, RF design, Transformer design, Bench instrumentation, CAD (SolidWorks), Prototyping, Drone hardware design, Arduino
Software: C, C++, Python, RISC-V assembly, Java, SQL, Scheme

Experience

Engineering Solutions at Berkeley Sept 2025 – Present

Electrical and Software Engineering Consultant *Berkeley, CA*

- Selected into UC Berkeley's first student-run engineering consulting organization (competitive, all-majors application) to take on a client-sponsored hardware project, delivering a fully documented, client-ready solution
- In charge of the electrical and embedded-software scope on a small cross-disciplinary team, collaborating with other engineers to progress through semester-long projects following ESB's full consulting workflow from scoping through client handoff

Fusion Dynamics June 2026 – Aug 2026

Electrical Engineering Intern *Oakland, CA*

- Scoped a drone-agnostic autonomous docking station that concentrates landing intelligence in the dock rather than the aircraft, evaluating fiducial-marker landing, IR beacon tracking, and Betaflight command uplink
- Designed a four-LED 850 nm IR beacon board in KiCad, computationally validating candidate emitter geometries for rotational distinctness across the full 360 degrees of rotation so the pose solver could not lock onto a wrong orientation
- Assisted in designing a 4-layer 95 x 55 mm structural frame PCB serving as both the top plate and the power distribution board of a sub-250 g drone, consolidating mechanical mounting and current paths into a single part

UC Davis Undergraduate Level Research Program July 2024 – Aug 2024

Researcher – Physics & Engineering Cluster *Davis, CA*

- Applied nuclear and electrical engineering principles to analyze real-world systems, performing quantitative evaluations of fission/fusion processes, critical mass requirements, reactor constraints, and electro-optical behavior
- Executed laboratory investigations using interferometry, laser systems, optical waveguides, and fiber-optic setups to characterize electromagnetic propagation, assess signal transmission behavior, and interpret experimental results

Projects

RF Generator for Quadrupole Ion Guide | *Evan R. Williams Research Group (via ESB)* Feb 2026 – Present

- Designing a custom high-voltage RF generator for a charge-detection mass spectrometer, targeting 1 to 1.5 kVpp quadrupole and/or 250 Vpp ion funnel drives from 10 kHz to 1 MHz into 300 pF to 1 nF capacitive loads for precise ion confinement
- Currently designing the high-voltage resonator PCB, splitting the band between a 1:9 broadband transformer below 40 kHz and an LC tank above it, tuned by a trim capacitor and 16 relay-switched binary-weighted toroids (4 µH to 128 mH)
- Custom-wound and bench-tested transformer prototypes on Material 43 NiZn toroids and Material 77 MnZn U-cores, selecting a toroid core that delivered 474% higher step-up gain at 1 MHz than the U-core under identical 1:4 winding
- Researched and benchmarked 5+ commercial step-up transformers across multiple manufacturers, evaluating bandwidth (previously 100 Hz to 10 MHz), voltage gain, and power handling for high-voltage RF drive

Precision Focus Actuator | *Malin Space Science Systems (via ESB)* Sept 2025 – Dec 2025

- Designed and prototyped a focus actuator for space camera applications used by NASA rovers and other commercial aerospace customers, requiring +/- 2 mil actuation precision within a tightly constrained mechanical envelope
- Designed a bidirectional homing and midpoint-referencing algorithm using a DRV8834 microstepped motor, dual limit switches, and a hall sensor to compensate for mechanical backlash, enabling 0.139 mil actuation per degree of rotation
- Validated positional accuracy and repeatability across 3D-printed iterations through step–displacement calibration, hall sensor median filtering tests, limit switch reliability checks, and closed-loop backlash compensation verification